

Winning Space Race with Data Science

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Introduction

- This presentation provides a concise summary of the detailed analyses conducted in the full notebooks. It highlights the key steps and findings, including data collection, preprocessing, exploratory analysis, visualization, and machine learning model development. The goal is to present the main insights and results in a clear and accessible format without delving into the extensive code and intermediate outputs found in the notebooks.
- All the details are available in the notebooks.

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- We began with Exploratory Data Analysis (EDA) and created a target label column called Class.
- The data was then standardized using StandardScaler, and split into training and testing sets with an 80/20 ratio. Four machine learning models were developed and evaluated: Logistic Regression, Support Vector Machine (SVM), Classification Trees, and K-nearest neighbour (KNN) For each model, GridSearchCV with 10-fold cross-validation was used to determine the best hyperparameters.
- The classification accuracy on the test set for each model was as follows:
 - Logistic Regression: 83.33%,
 - SVM: 83.33%
 - KNN: 83.33%
 - Classification Tree: 88%
- Logistic Regression, SVM, and KNN performed equally well, each achieving an accuracy of 83.33%, while the classification tree performed the best based on accuracy.

Background (Storytelling)

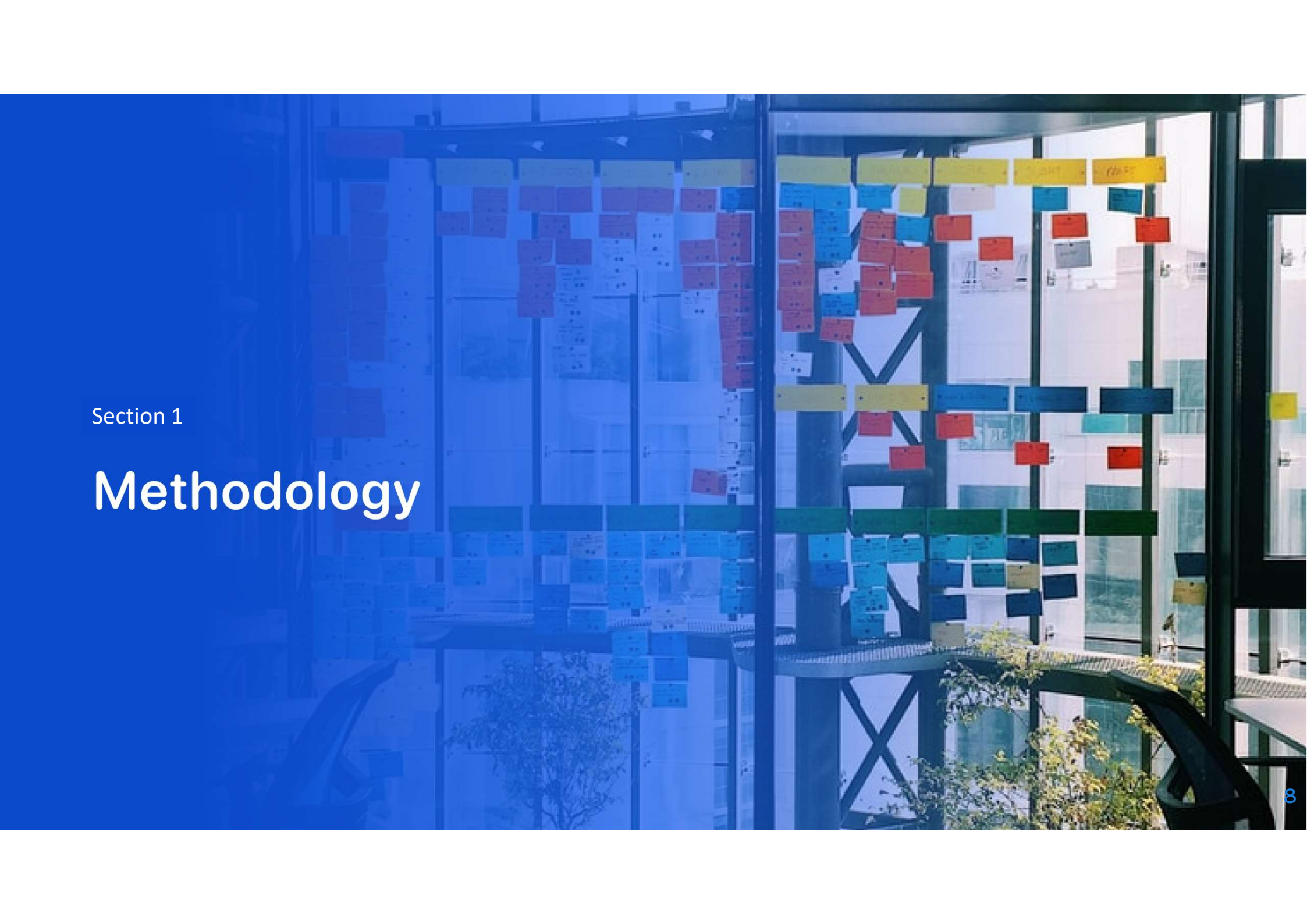
- Space Y is preparing to enter the commercial spaceflight industry and aims to compete with established players like SpaceX. One of the key innovations that has given SpaceX a competitive edge is its ability to reliably land and reuse the first stage of the Falcon 9 rocket, significantly reducing launch costs.
- While SpaceX offers launches for approximately \$62 million, other providers charge upwards of \$165 million — largely due to the inability to reuse rocket components.

Background

- To better understand and evaluate this competitive advantage, Space Y is conducting a predictive analysis based on publicly available SpaceX launch data.
- By building machine learning models that can forecast the success of a Falcon 9 first-stage landing, Space Y can gain insights into the operational efficiency, risks, and cost-saving potential of reusable launch systems.

The Goal

- The central question of this project is:
- Can we accurately predict whether the Falcon 9 first stage will land successfully using historical launch data?



Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - SpaceX launch data from a public API
 - Falcon 9's Wikipedia page till 21st of June 2021
- Perform data wrangling
 - Data was cleaned, missing values were filled and necessary features were engineered
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium
- Perform predictive analysis using classification models

Data Collection

- Data for this project was gathered from two main sources:
 - SpaceX launch data retrieved via a public API, providing detailed information on Falcon 9 missions.
 - Supplementary information from Falcon 9's Wikipedia page, with data current up to June 21, 2021.
- These sources provided comprehensive records on launch outcomes, vehicle configurations, and mission details essential for building predictive models.

Data Collection

Flowchart of data collection:

- Identify relevant data sources
- Access public API for SpaceX launch data
- Extract Falcon 9 mission details
- Collect supplementary data from Wikipedia (up to June 21, 2021)
- Merge API and Wikipedia datasets
- Clean and validate combined dataset
- Prepare dataset for analysis and modelling

Data Collection – SpaceX API

- Request library was used to fetch the data from the API and store in a Pandas' data frame

```
response.content[:1000]
```

```
b'[{{"fairings":{"reused":false,"recovery_attempt":false,"recovered":false,"ships":[]},"links":{"patch":{"small":"https://images2.imgbox.com/94/f2/NN6Ph45r_o.png","large":"https://images2.imgbox.com/5b/02/QcxHUb5V_o.png"},"reddit":{"campaign":null,"launch":null,"media":null,"recovery":null},"flickr":{"small":[],"original":[]},"presskit":null,"webcast":"https://www.youtube.com/watch?v=0a_00nJ_Y88","youtube_id":"0a_00nJ_Y88","article":"https://www.space.com/2196-spacex-inaugural-falcon-1-rocket-lost-launch.html","wikipedia":"https://en.wikipedia.org/wiki/DemoSat"},"static_fire_date_utc":"2006-03-17T00:00:00.000Z","static_fire_date_unix":1142553600,"net":false,"window":0,"rocket":"5e9d8d95eda69955f709d1eb","success":false,"failures":[{"time":33,"altitude":null,"reason":"merlin engine failure"}],"details":"Engine failure at 33 seconds and loss of vehicle","crew":[],"ships":[],"capsules":[],"payloads":["5eb0e4b5b6c3bb006eeb1e1"],"launchpad":"5e9e4502f5090995de566f86","flight_number":1,"name":'}
```

Next, let's try to discover some more relevant information for this project.

Request and parse the SpaceX launch data using the GET request

To ensure consistent results throughout this project, we will use a static JSON response object. This means that running the notebooks multiple times will produce the same outputs, making the analysis reproducible and the results reliable.

```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'
```

We should see that the request was successful with the 200 status response code

```
response=requests.get(static_json_url)
```

```
response.status_code
```

```
200
```

Now we decode the response content as a JSON using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
data=response.json()
data= pd.json_normalize(data)
```

Data Collection - Webscraping

- BeautifulSoup library was used to scrape the data from the Wikipedia page of Falcon 9

To ensure consistent results throughout this project, we will use a static JSON response object. This means that running the notebooks multiple times will produce the same outputs, making the analysis reproducible and the results reliable.

```
: static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
    "AppleWebKit/537.36 (KHTML, like Gecko) "
    "Chrome/91.0.4472.124 Safari/537.36"
}
```

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

```
: response=requests.get(static_url, headers=headers)
```

```
: response
```

```
: <Response [200]>
```

We will create BeautifulSoup object from the HTML response

```
: soup=BeautifulSoup(response.text, "html.parser")
```

```
: print(soup.title)
```

```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

Data Wrangling

- After collecting data from the API and Wikipedia page, several columns were deemed irrelevant for the predictive model, including those containing links to launch footage on various social media platforms.

```
data.head()
```

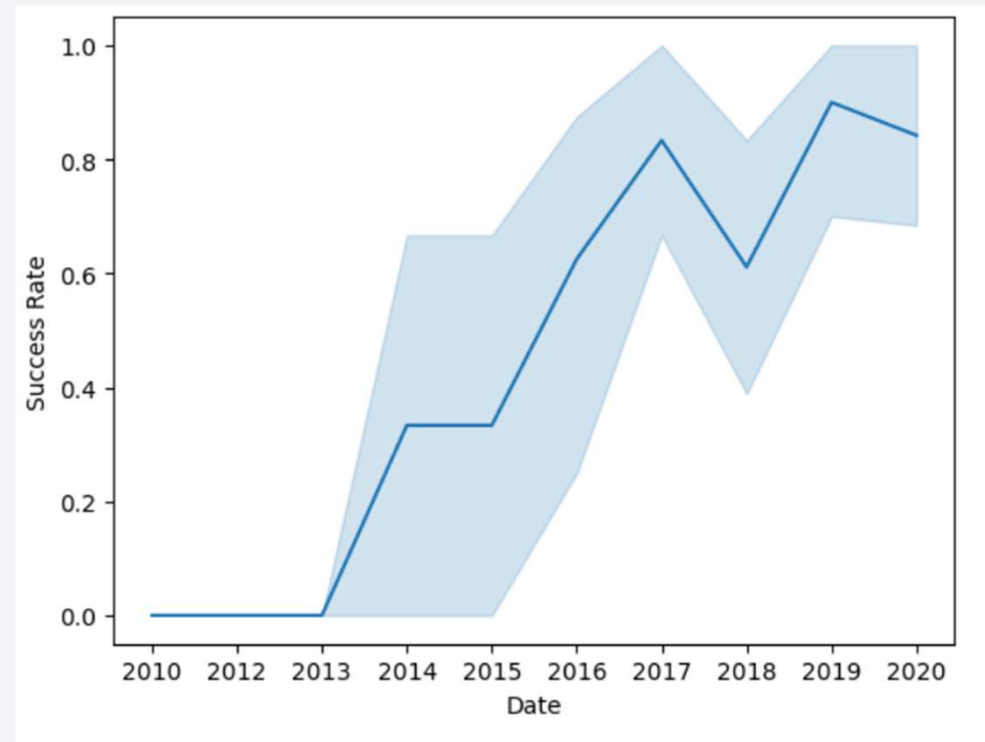
s.ships	links.patch.small	links.patch.large	links.reddit.campaign	links.reddit.launch	links.reddit.media	links.reddit.recovery	links.flickr.small	links.flickr.o
	https:// images2.imgbox.com/3c/0e/ T8jlCN3_o.png	https:// images2.imgbox.com/40/ e3/GypSkayF_o.png	None	None	None	None		
	https:// images2.imgbox.com/4f/e3/ I0lkuj2e_o.png	https:// images2.imgbox.com/be/ e7/iNqsqVYM_o.png	None	None	None	None		

Data Wrangling

- The original dataset's success summary included various types of information, such as the timing of the landing and whether it occurred on a sea platform or land.
- To streamline the data, a new feature was engineered to indicate the success or failure of each landing. This was done by consolidating all successful landings into a single column called "Class," which holds two values: 1 for successful landings and 0 for unsuccessful ones.

EDA with Data Visualization

- To better understand the relationships between the various parameters, the data was visualized using a variety of methods, including categorical plots, scatter plots, bar charts, and line plots.
- For example, a line plot was used to visualize the mission success rate over the years, highlighting trends and changes in performance over time.



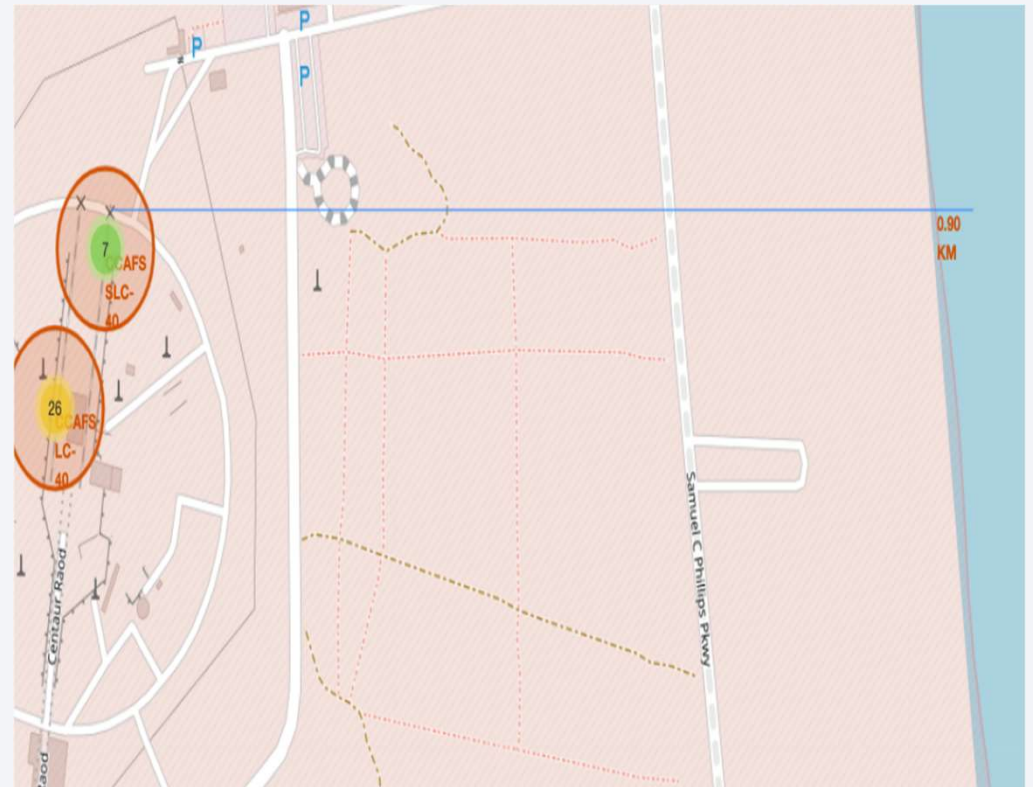
EDA with SQL

A SQLite database was used to further explore and understand the dataset, and the following queries were performed:

- Retrieve all unique launch sites in the dataset.
- Display five records where the launch site names start with “CCA”.
- Calculate the total payload mass for missions conducted for the customer “NASA (CRS)”.
- Determine the average payload mass for the booster version “F9 v1.1”.
- Find the earliest date on which a successful ground pad landing occurred.
- List booster versions that achieved successful drone ship landings with payload masses between 4,000 kg and 6,000 kg.
- Count the total number of mission outcomes grouped by success and failure.
- Identify booster versions that carried the maximum payload mass using a subquery.
- List records from 2015 with failed drone ship landings, including the month, booster version, launch site, and landing outcome.
- Rank landing outcome counts in descending order for missions conducted between 2010-06-04 and 2017-03-20.

Build an Interactive Map with Folium

- Markers, circles, and lines were added to the Folium map to identify launch sites, visualize their surrounding proximity areas, and display distances between locations.
- This helped reveal geographical patterns that may influence launch success.



Predictive Analysis (Classification)

- The process began with Exploratory Data Analysis (EDA), during which a target label column called Class was created.
- The data was then standardized using StandardScaler and split into training and testing sets with an 80/20 ratio.

Predictive Analysis (Classification)

- Four machine learning models were developed and evaluated: Logistic Regression, Support Vector Machine (SVM), Classification Trees, and K-Nearest Neighbors (KNN).
- For each model, GridSearchCV with 10-fold cross-validation was applied to identify the optimal hyperparameters.
- Model performance was evaluated using accuracy scores, and confusion matrices.

Results

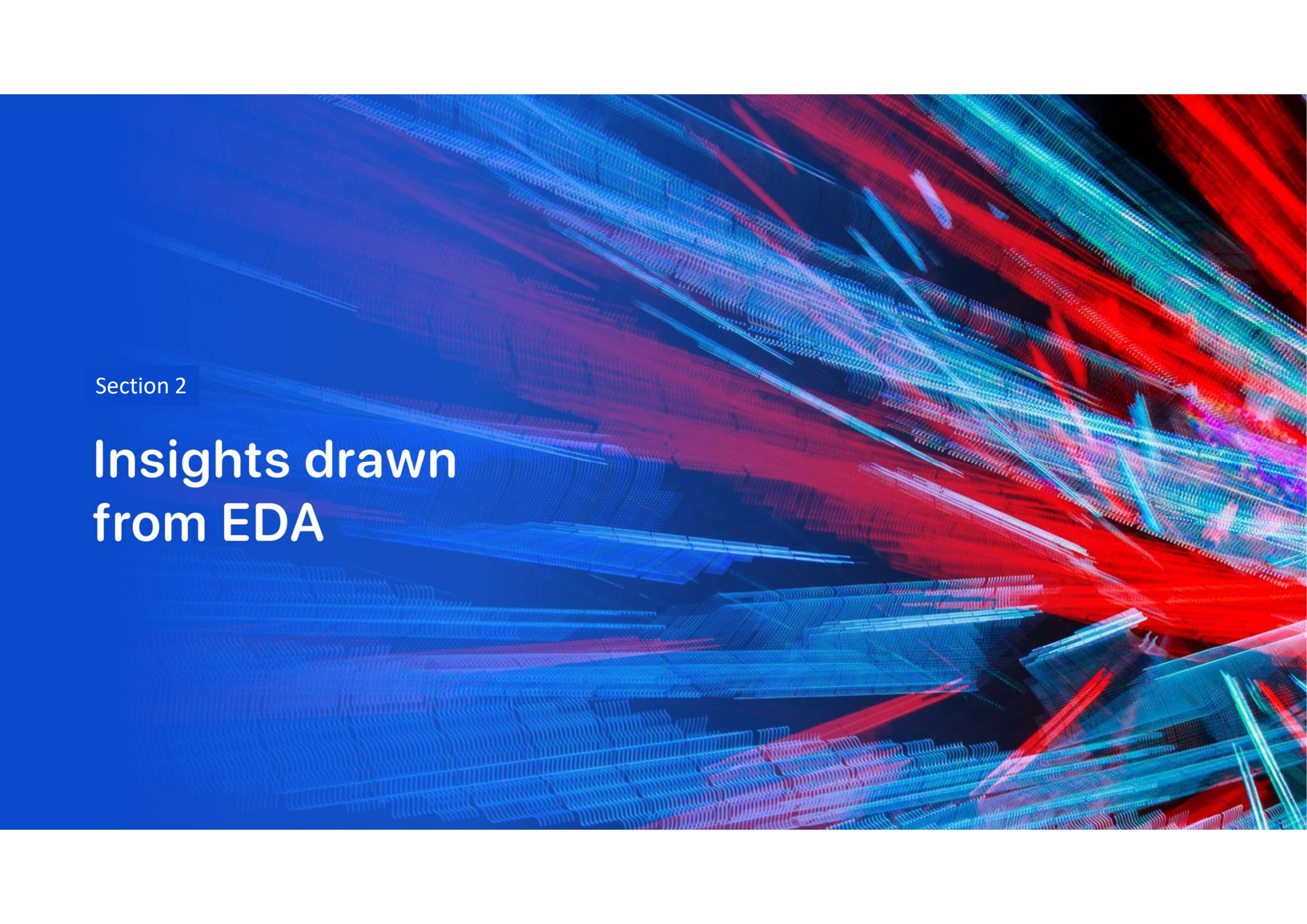
Exploratory Data Analysis revealed several key insights:

- Landing success improved as the flight number increased.
- The majority of launches occurred at CCAFS SLC-40, followed by KSC LC-39A, with the fewest launches taking place at VAFB SLC-4E.
- Orbits such as ES_L1, GEO, HEO, and SSO demonstrated a perfect 100% success rate.
- Heavy payloads were observed to have a negative impact on GTO orbits, but a positive effect on Polar LEO and ISS orbits.
- Overall, the mission success rate steadily increased from 2013 to 2020.

Results

The classification accuracy on the test set for each model was:

- Logistic Regression: 83.33%
- Support Vector Machine (SVM): 83.33%
- K-Nearest Neighbors (KNN): 83.33%
- Classification Tree: 88%

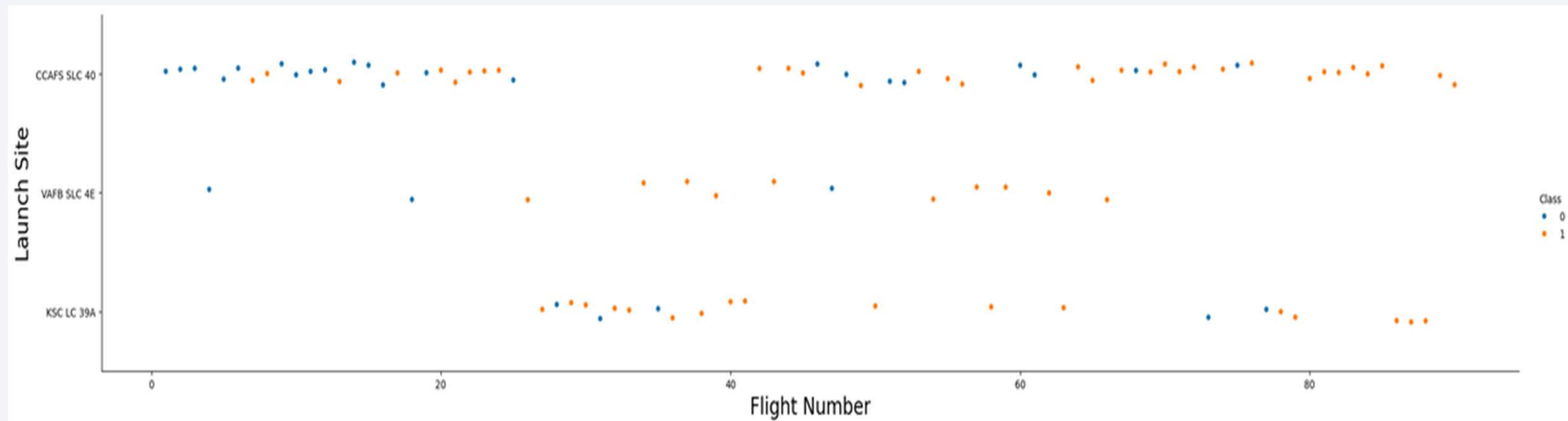
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and movement, reminiscent of a digital or data visualization theme.

Section 2

Insights drawn from EDA

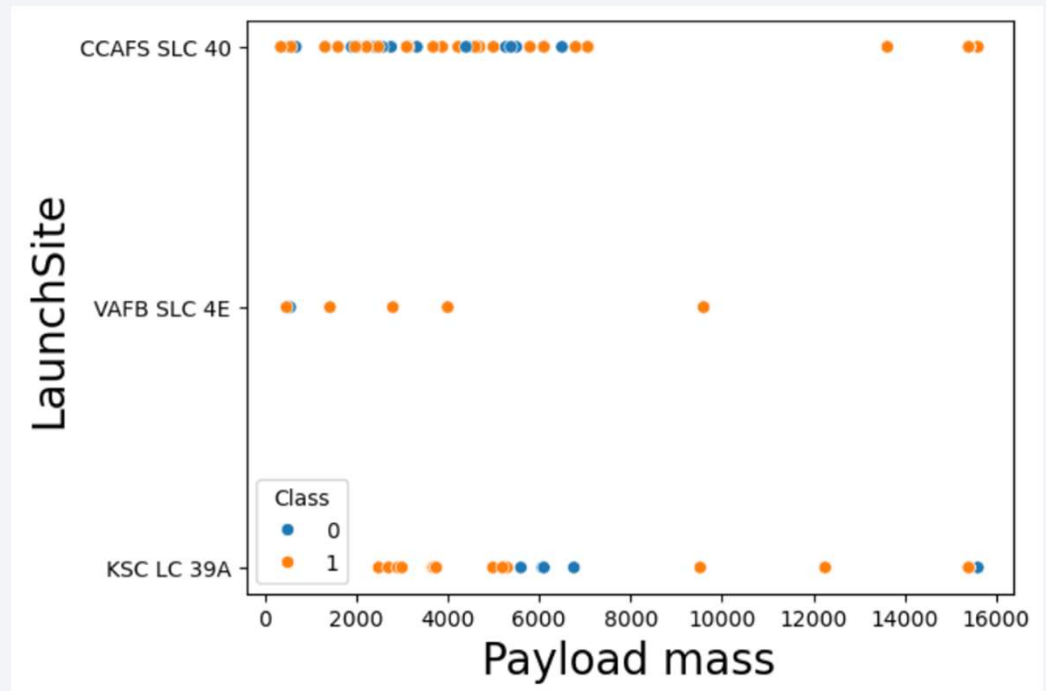
Flight Number vs. Launch Site

- Landing success increased as the flight number rose, indicating improved performance with experience.



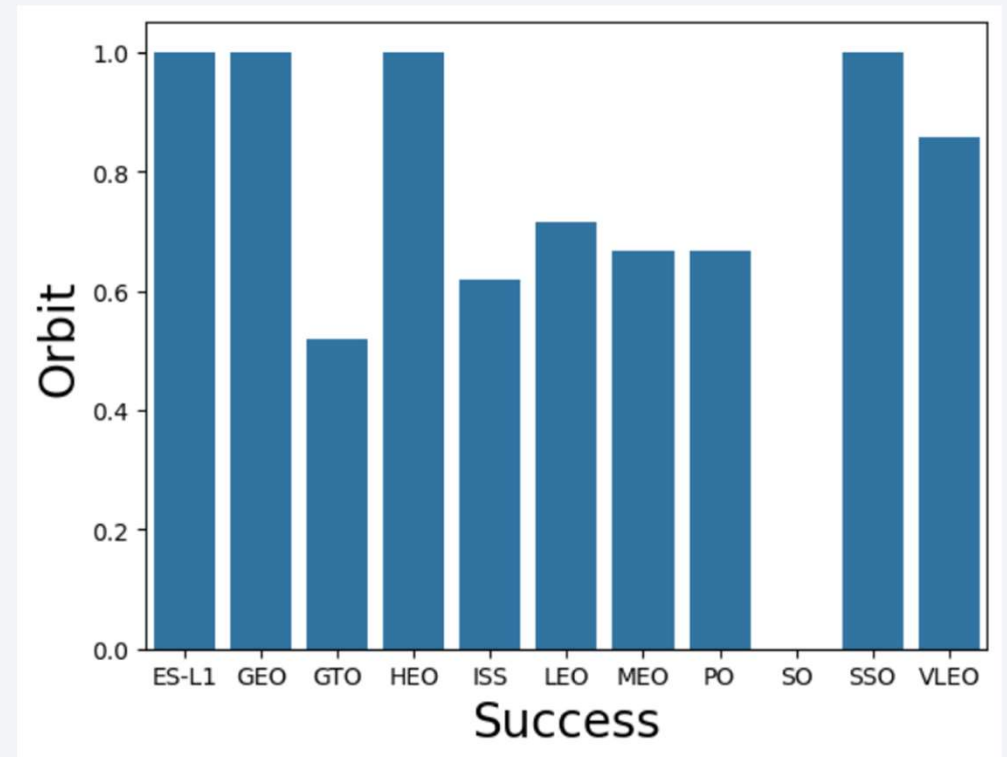
Payload vs. Launch Site

- From the Payload Mass vs. Launch Site scatter plot, it can be observed that no rockets carrying heavy payloads (greater than 10,000 kg) were launched from the VAFB-SLC site.



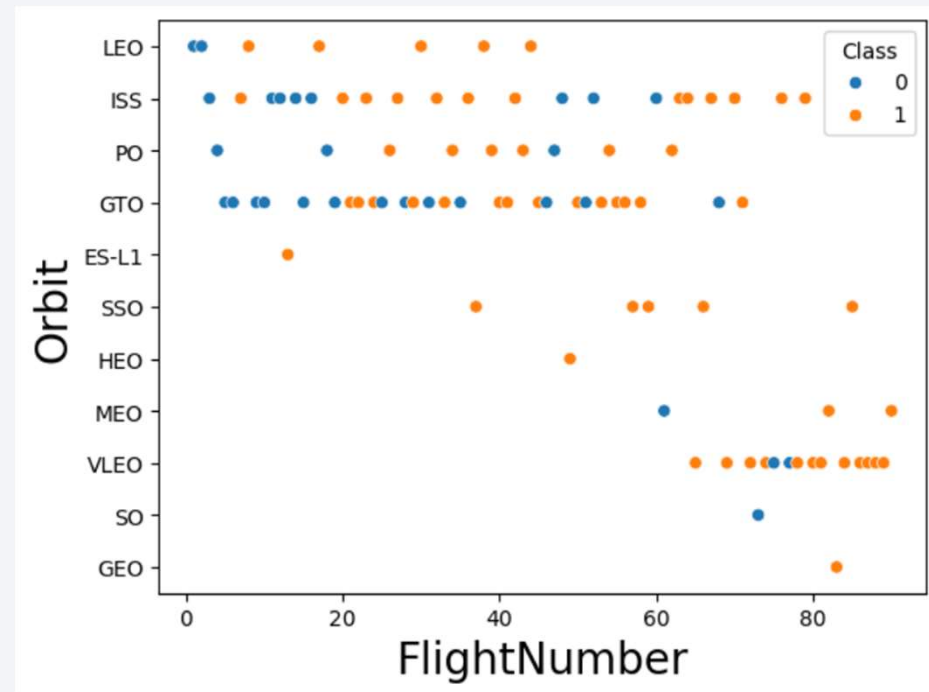
Success Rate vs. Orbit Type

- The ES_L1, GEO, HEO, and SSO orbits each achieved a perfect 100% success rate.



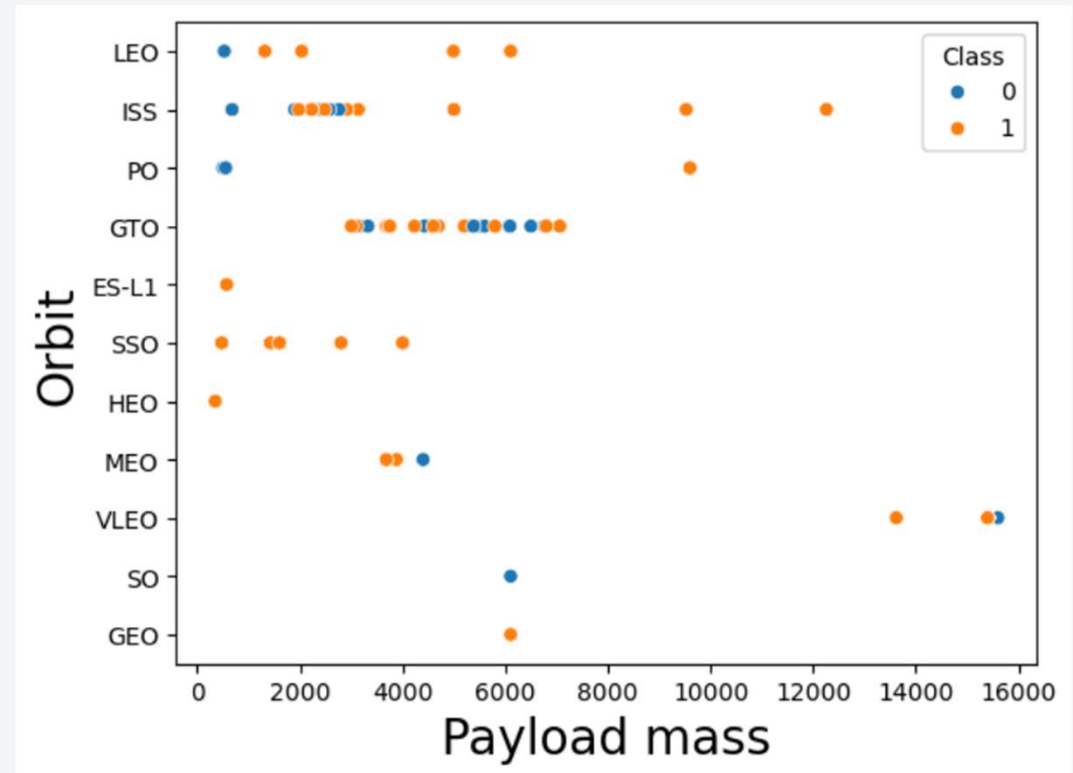
Flight Number vs. Orbit Type

- In the LEO orbit, success appears to improve with an increasing number of flights, while in the GTO orbit, there is no clear relationship between flight number and success.



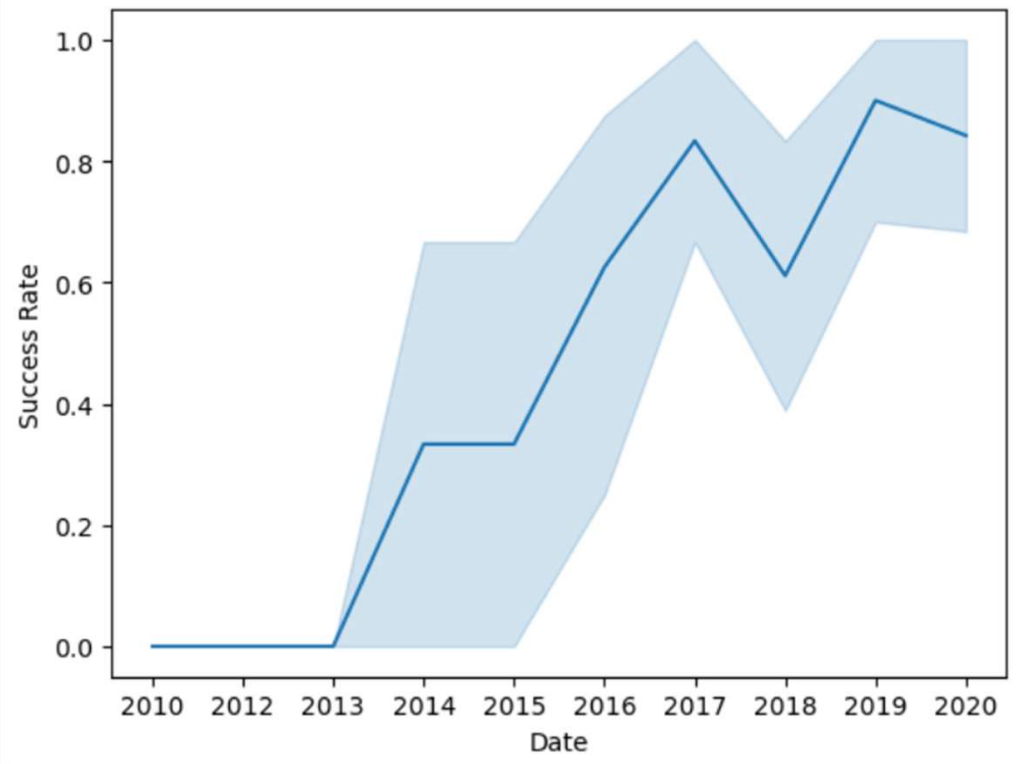
Payload vs. Orbit Type

- For heavy payloads, successful landings are more frequent in Polar, LEO, and ISS orbits.
- In contrast, for GTO orbits, successful and unsuccessful landings occur at similar rates, making it difficult to identify a clear trend.



Launch Success Yearly Trend

- The mission success rate steadily increased from 2013 to 2020.



SQL: All Launch Site Names

Unique Launch Site Names:

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

- These represent the distinct locations from which SpaceX has conducted rocket launches, as recorded in the dataset.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

SQL: Launch Site Names Begin with 'CCA'

- The following SQL result table shows the first five launches from sites starting with "CCA" (Cape Canaveral Air Force Station), including details such as launch dates, booster versions, payloads, orbits, customers, and mission outcomes.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outc
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

SQL: Total Payload Mass

- The total payload mass carried by boosters for NASA missions, calculated by summing the PAYLOAD_MASS_KG_ for all records where the customer is NASA, is 45,596 kilograms.

SUM(PAYLOAD_MASS_KG_)
45596

SQL: Average Payload Mass by F9 v1.1

- The average payload mass carried by the F9 v1.1 booster version is 2928.4 kg, calculated by averaging the `PAYLOAD\_MASS\_KG` values for all launches using this booster type.

avg(PAYLOAD_MASS_KG)
2928.4

SQL: First Successful Ground Landing Date

- This date represents the earliest successful landing on a ground pad, determined by filtering records where the Landing_Outcome is "Success (ground pad)" and selecting the minimum Date value.

min(Date)
2015-12-22

SQL: Successful Drone Ship Landing with Payload between 4000 and 6000

- These booster versions successfully landed on a drone ship and carried payloads within the specified mass range (4000–6000 kg), as filtered using the Landing_Outcome and PAYLOAD_MASS__KG_ columns.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

SQL: Total Number of Successful and Failure Mission Outcomes

Mission Outcome Counts:

- Failure (in flight): 1
- Success: 99 (including the duplicate success label)
- Success (payload status unclear): 1

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

SQL: Boosters Carried Maximum Payload

- These boosters carried the maximum recorded payload mass as identified using a subquery that selects boosters matching the maximum PAYLOAD_MASS__KG_ value in the dataset.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

SQL: Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- This query counts and ranks different landing outcomes that occurred between the specified dates, showing "No attempt" as the most common outcome, followed by successful and failed landings on drone ships.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

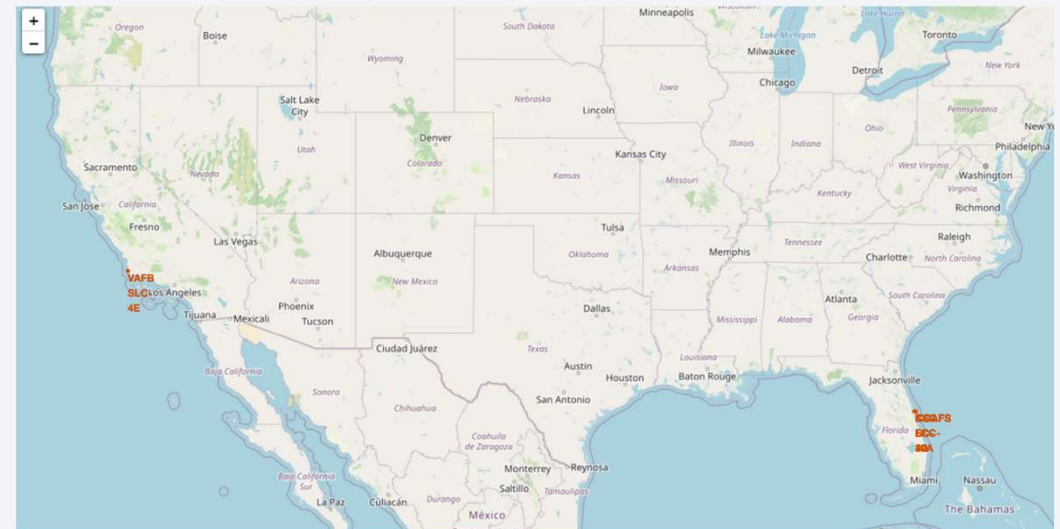
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is used as a background for the title slide.

Section 3

Launch Sites Proximities Analysis

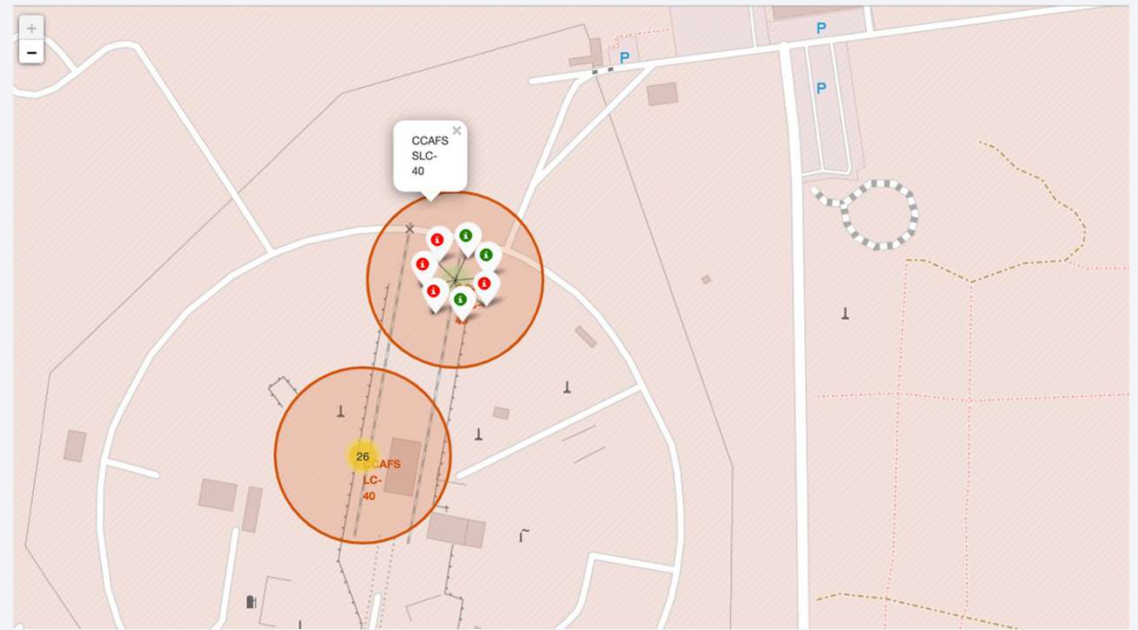
Folium Launch Sites

- The folium map shows SpaceX launch sites with most clustered in Florida for ISS and GTO missions, and one in California for polar orbits, highlighting Florida's strategic launch advantage.



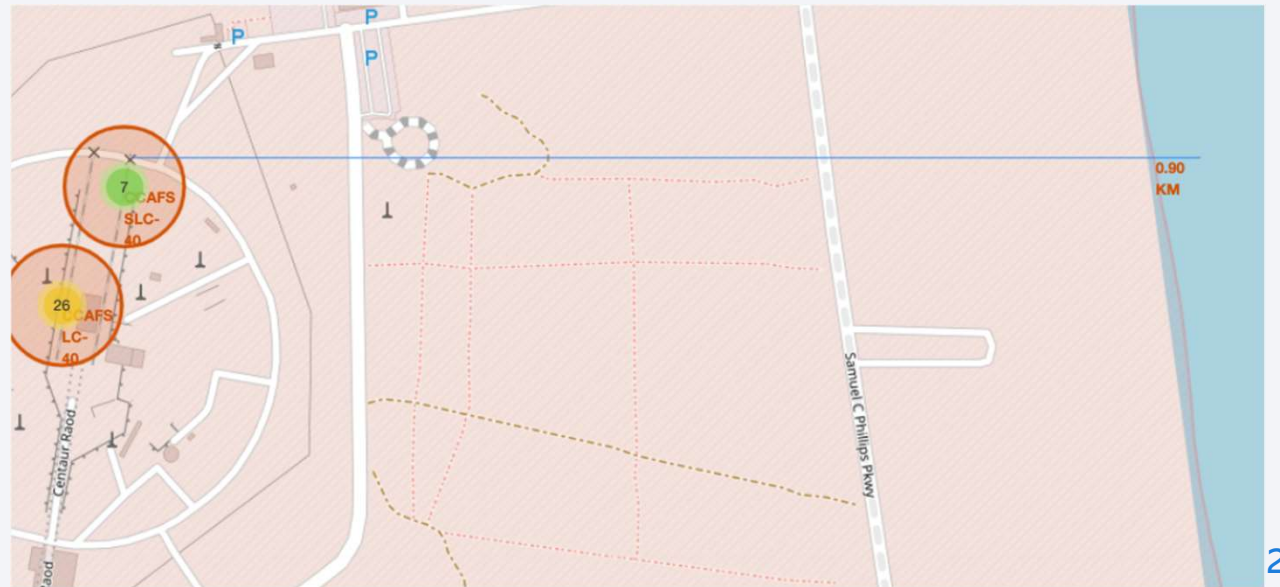
Folium Labels

- The color-coded markers in the Folium marker clusters visually differentiate launch sites, making it easy to identify sites with higher success rates by the prevalence of success-colored markers.



Folium Proximity

- By examining all the folium maps, we can conclude that launch sites are located near coasts, close to major highways and railways for logistics, and away from densely populated urban areas.





Section 4

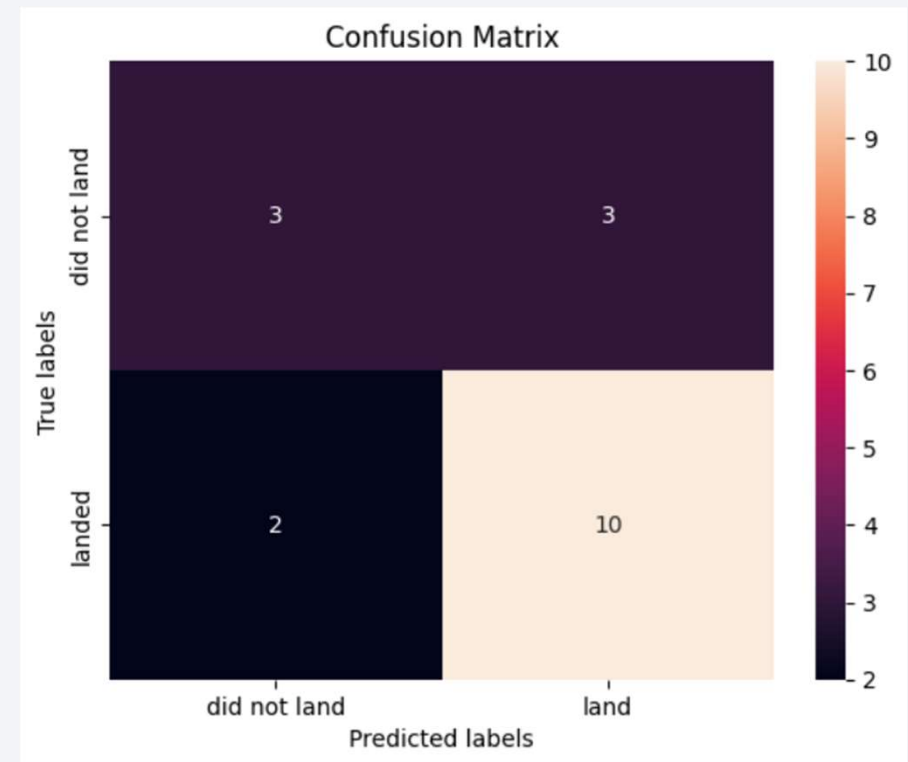
Predictive Analysis (Classification)

Classification Accuracy

- The model with the highest classification accuracy was the classification tree, achieving approximately 88% accuracy. This indicates that classification tree was the most effective among the tested models at predicting whether the Falcon 9 first stage would land successfully, demonstrating strong capability in capturing the complex patterns within the data.

Confusion Matrix

- The model performs very well in distinguishing between successful and unsuccessful landings, correctly classifying most cases.
- Only one misclassification occurred for each class, indicating minimal confusion.
- This strong classification capability is consistent with the high overall accuracy of 88%, reflecting the model's reliability in predicting landing outcomes.



Conclusions

- Data Collection: Collected SpaceX launch data including launch sites, payloads, booster versions, orbits, mission outcomes, and landing results using the SpaceX API and web scraping from the Falcon 9 Wiki.
- Data Wrangling: Cleaned and prepared the data by creating target labels, standardizing features, and splitting the data into training and testing sets for model building.

Conclusions

- Data Visualization: Used various charts like categorical plots, scatterplots, bar charts, and line graphs to explore and understand the relationships between different features.
- Model Performance: Among the machine learning models tested, the classification tree model performed best with about 88% accuracy; the confusion matrix showed it predicted well overall but made some errors in one class.

Thank you!

